Lecture 4 · Powers, logarithms, growth and target times

20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer; partial completion is welcome.

Q1 · Foundation

Without using your calculator, evaluate 8^2.

A. 16

B. 64

C. 10

D. 256

Source: Jacques 9e, §2.3, Exercise 2.3, 1(1)(a); printed p. 168 / PDF p. 185.

Q2 · Foundation

Without using your calculator, evaluate 3^(-1).

A. -3

B. 3

C. 1/3

D. -1/3

Source: Jacques 9e, §2.3, Exercise 2.3, 1(1)(c); printed p. 168 / PDF p. 185.

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Q3 · Foundation

Without using your calculator, evaluate 36^(1/2).

A. 18

B. 72

C. 1296

D. 6

Source: Jacques 9e, §2.3, Exercise 2.3, 1(1)(f); printed p. 168 / PDF p. 185.

Q4 · Core

Use the rules of indices to simplify $a^3 \times a^{(8)}$.

A. a^(11)

B. a^(24)

C. 2a^(11)

D. a^(5)

Source: Jacques 9e, §2.3, Exercise 2.3, 2(a); printed p. 168 / PDF p. 185.

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Q5 · Core

Use the rules of logs to expand ln(xy). Assume x>0 and y>0.

A. ln x ln y

B. ln x+ln y

C. ln x-ln y

D. x+y

Source: Jacques 9e, §2.4, Exercise 2.4, 2(a); printed p. 183 / PDF p. 200.

Q6 · Core

Solve e^(x)=5.9. Round your answer to two decimal places.

A. 5.90

B. 0.59

C. 1.77

D. -1.77

Source: Jacques 9e, §2.4, Exercise 2.4, 4(a); printed p. 183 / PDF p. 200.

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Q7 · Application

A teaching growth model is N(t)=100(1.20)^(t), with t measured in weeks. What does the model predict at t=2?

A. 120

B. 140

C. 240

D. 144

Source: Jacques 9e, §2.3, Original supplementary classroom model, Lecture 4: N(t)=100(1.20)^t; printed p. 151 / PDF p. 168.

Q8 · Application

For N(t)=100(1.20)^(t), the continuous model reaches 200 at approximately 3.80 weeks. If reports are made only at whole weeks, which is the FIRST report at or above 200?

A. Week 4

B. Week 3

C. Week 3.80

D. Week 5

Source: Jacques 9e, §2.4, Original supplementary target-time application, Lecture 4: continuous crossing and whole-week reporting; printed p. 170 / PDF p. 187.

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Q9 · Application

Solve ln x=0. Which x intercept does this give for y=ln x?

A. x=0

B. x=1

C. x=e

D. x=-1

Source: Jacques 9e, §2.4, Exercise 2.4, 6(b); printed p. 183 / PDF p. 200.

Q10 · Challenge (optional)

Future sales of two products A and B are given by S_A=5e^(0.01t) and S_B=2e^(0.02t). Find the time, t, when sales of the two products are the same.

A. ln(2.5)

B. 100ln(0.4)

C. 100ln(2.5)

D. 50ln(2.5)

Source: Jacques 9e, §2.4, Exercise 2.4, 7; printed p. 183 / PDF p. 200.